

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 24

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
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| 01.1     | • $15 - (-18) = 33\text{ }^{\circ}\text{C}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1     |
| 01.2     | <ul style="list-style-type: none"> <li>• Visible light passes easily through the atmosphere and is absorbed by the surface, which warms</li> <li>• The warm surface emits (long-wave / far) infrared radiation</li> <li>• Greenhouse gases (water vapour, <math>\text{CO}_2</math>, methane) absorb infrared better than the average for the atmosphere and convert it to heat</li> <li>• The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction</li> </ul> | 4     |
| 01.3     | • Any gas that is better at absorbing infrared radiation than the average for the atmosphere                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| 01.4     | <ul style="list-style-type: none"> <li>• Arriving: mainly ultraviolet, visible light and near infrared (short wavelength); leaving: mainly long-wavelength (far) infrared</li> <li>• Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained</li> </ul>                                                                                                                                                            | 2     |
| 01.5     | <ul style="list-style-type: none"> <li>• The energy and the processes it drives control climate, ocean currents and the hydrological cycle</li> <li>• These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive</li> </ul>                                                                                                                                                                                                                         | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.                                                                                                                                                                                                                                                                                                                                                                                                                 |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| 02.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO<sub>2</sub> and CH<sub>4</sub>, climate change, ocean acidification, reduced soil organic matter</li> <li>• Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO<sub>x</sub>; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N<sub>2</sub>O warming, NO<sub>x</sub> toxicity</li> <li>• Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage)</li> <li>• Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones</li> <li>• Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement</li> </ul> | 9     |
| 02.2     | <ul style="list-style-type: none"> <li>• The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 148-157; AQA spec 3.2.4.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| 03.1     | <ul style="list-style-type: none"> <li>• B - rivers are accessible over a large area with few natural contaminants</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1     |
| 03.2     | <ul style="list-style-type: none"> <li>• Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health)</li> <li>• Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
| 03.3     | <ul style="list-style-type: none"> <li>• Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers)</li> <li>• Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>04.1</b> | • Organic (methyl) mercury                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     |
| <b>04.2</b> | • Anaerobic microbes converted the inorganic mercury into organic methyl mercury<br>• Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people                                                                                                                                                                                                                                                                       | 2     |
| <b>04.3</b> | • Most heavy metals are less soluble at high pH<br>• So they remain immobile and are not leached into water or absorbed by organisms                                                                                                                                                                                                                                                                                                                                                              | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.                                                                                                                                                                                                                                                                                                                                                                        |       |
| <b>05.1</b> | • Gasification: coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane)<br>• Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS<br>• Liquefaction: coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion<br>• Provides liquid fuels for vehicles / aircraft that solid coal cannot supply | 4     |
| <b>05.2</b> | • A solid, ice-like crystalline structure in which methane is trapped within water ice<br>• Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins                                                                                                                                                                                                                                                                                 | 2     |
| <b>05.3</b> | • Water heating: hot water pumped into sediments melts the hydrate releasing methane<br>• Depressurisation: drilling reduces pressure so methane dissociates from the crystals<br>• CO <sub>2</sub> injection: CO <sub>2</sub> bonds more strongly to the ice and displaces methane (could double as CCS)<br>• Risks: methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback; seabed ecology poorly understood; sediment instability / landslides          | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                          |       |
| <b>06.1</b> | • About 8 nights<br>• The running mean fluctuates widely with few samples but stabilises at 21-22 from 8 nights onwards; extra samples do not change it significantly                                                                                                                                                                                                                                                                                                                             | 2     |
| <b>06.2</b> | • Moth activity varies with temperature, wind, rain and season (adults only fly for part of the year) so catches vary independently of population size<br>• Sampling at different conditions / times would make results not comparable<br>• Timing between samples should be chosen to detect the temporal variation of interest (eg weekly through the season)                                                                                                                                   | 3     |
| <b>06.3</b> | • Accuracy: how close a recorded result is to the real value<br>• Precision: the smallest interval between possible recorded values (eg a ruler marked in mm is more precise than one in cm)                                                                                                                                                                                                                                                                                                      | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| <b>06.4</b> | <ul style="list-style-type: none"> <li>• A null hypothesis (eg no difference / no correlation) can be tested statistically and accepted or rejected at a stated confidence level</li> <li>• Standardised techniques ensure results collected at different times, places or by different people are comparable</li> <li>• Other variables are controlled or monitored so their effect can be assessed; allows the study to be repeated / checked (reliability)</li> </ul>                                                                                                                           | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| <b>07.1</b> | <ul style="list-style-type: none"> <li>• Most energy comes from non-renewable fossil fuels whose reserves are depleting; the rate of new discoveries has fallen sharply since the 1980s</li> <li>• Use causes unacceptable environmental damage: atmospheric pollution, climate change, oil pollution, habitat loss</li> <li>• Economic / social: depleting resources become more expensive, reducing future affluence; risk of an energy gap if new technologies are not developed in time</li> </ul>                                                                                             | 3     |
| <b>07.2</b> | <ul style="list-style-type: none"> <li>• Coal mining: habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise</li> <li>• Oil and gas extraction: spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO<sub>2</sub> / CO<sub>2</sub> / smoke</li> <li>• Pipelines causing habitat damage during construction; tanker transport risks</li> <li>• Fuel processing (refineries, coal washing) pollution; embodied energy of equipment and site development for all resources</li> </ul>                    | 4     |
| <b>07.3</b> | <ul style="list-style-type: none"> <li>• New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles)</li> <li>• Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise</li> <li>• By then depletion would reduce supply faster than replacements could be built, causing shortages</li> </ul>                                                                                                                                                                                         | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| <b>08.1</b> | <ul style="list-style-type: none"> <li>• Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960)</li> <li>• Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level)</li> </ul>                                                                                                                                                                                                                                                                                                                     | 2     |
| <b>08.2</b> | <ul style="list-style-type: none"> <li>• <math>(200 - 25) / 200 \times 100</math></li> <li>• = 87.5%</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| <b>08.3</b> | <p>Any 4 from:</p> <ul style="list-style-type: none"> <li>• DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator)</li> <li>• Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths</li> <li>• Higher doses caused sterility / death</li> <li>• After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery)</li> <li>• Breeding success recovered so the population rose</li> </ul> | 4     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
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| <b>08.4</b> | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• DDT persists in soil / sediments / food chains for many years so exposure continued</li> <li>• Birds of prey are k-selected: few young per year, slow to mature so populations grow slowly</li> <li>• Other pressures (habitat loss, persecution)</li> </ul>                                                                                                                                                                                         | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.                                                                                                                                                                                                                                                                                                                                                                                               |       |
| <b>09.1</b> | <ul style="list-style-type: none"> <li>• Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored</li> <li>• Hydrogen has a high energy density so it can be transported / stored for long periods</li> <li>• Released by combustion for heat, industrial processes or steam turbines; or by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water</li> <li>• Can also be used as a vehicle fuel or converted to methane / ammonia</li> </ul> | 4     |
| <b>09.2</b> | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks</li> <li>• Liquefied at very low temperature - energy for refrigeration and compression</li> <li>• Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy</li> <li>• Conversion to ammonia - some energy needed but less than the alternatives</li> </ul>                      | 3     |
| <b>09.3</b> | <ul style="list-style-type: none"> <li>• Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging</li> <li>• A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied</li> </ul>                                                                                                                                                                                             | 2     |
| <b>09.4</b> | <ul style="list-style-type: none"> <li>• Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity</li> </ul>                                                                                                                                                                                                                                                             | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| <b>10.1</b> | <ul style="list-style-type: none"> <li>• <math>4.2 / 0.7 = 6</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| <b>10.2</b> | <ul style="list-style-type: none"> <li>• Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low</li> <li>• In winter days are very short / the Sun does not rise, so light levels are extremely low</li> </ul>                                                                                                                                                                                                                                                 | 2     |
| <b>10.3</b> | <ul style="list-style-type: none"> <li>• Reflectors can only focus parallel rays of direct sunlight onto the absorber</li> <li>• Clouds, smoke and dust scatter the light so it cannot be focused and intensified</li> </ul>                                                                                                                                                                                                                                                                                            | 2     |
| <b>10.4</b> | <ul style="list-style-type: none"> <li>• Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further)</li> </ul>                                                                                                                                                                                                                                                                                                                         | 1     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
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| 10.5     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Problems: intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature</li> <li>• Photothermal: evacuated tubes reduce conduction / convection losses; thermal stores (water, sand, concrete, molten salt) allow later use; passive solar architecture, brise soleil, light tubes; heat pumps deliver up to four times the energy used</li> <li>• Photovoltaic developments: multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces</li> <li>• Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency</li> <li>• Integration: grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management</li> <li>• Evaluation: strong in sunny deserts / low latitudes, weaker in high latitudes in winter; cost falling; manufacturing impacts; no single technology solves all problems; justified conclusion</li> </ul> | 9     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |

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| 11.1     | <p><b>Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude)</li> <li>• Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use</li> <li>• New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning</li> <li>• HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities</li> <li>• Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources</li> <li>• Evaluation and justified conclusion on how far these resources can replace fossil fuels</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
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| 11.2     | <p><b>Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral</li> <li>• Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud</li> <li>• Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring</li> <li>• Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice</li> <li>• Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring</li> <li>• Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.</p> | 25    |